

BUTTON

MULTI-TOOL

USER MANUAL AND GUIDE

A/B MULTIMODES

A/B Mode: RAMPS up/dn
TOGGLE
Random CV (bi)
Random CV (uni)
Mersenne Twister (seeded random)
RAMPS up/dn ✓
RAMPS up/dn latching

UTILITY MODES

Utility: all TRIGGERS blocked by PROBABILITY
END OF GATE
START & END OF GATE
all TRIGGERS blocked by PROBABILITY ✓
A PLUS B
A MINUS B
A DIVED BY B
Initialize
Parameter name ▶
Set default to current value

BUTTON

MANUAL PUSH BUTTON

IN

in: GATE | CLOCK | TRIG

%

PROBABILITY global (0-100%)

TR

TRIG out

GT

GT out

GT length (0.2s-10s / 0=auto)

A

A out

A range/volt & A offset

B

B out

B range/volt & B offset

<RAMPS>

Multimode selector

Utility out

Utility mode

Intended as an advanced trigger and clock modulation Utility.
Over time it grew to 700 lines of code incorporating a variety of light-weight features. Driven by a single Clock/Trigger/Gate/CV input (polyphony summed through OR-Logic). The panel gives access to simple core functions with the advanced options nested into the context menu. Functions are isolated by design and operate without loops (hardware friendly).

Button for manually triggering the module, can be held to keep Gate (auto) high.

Input reacts to signals above 0.3 / below -0.3 volts and can be driven by

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bipolar signals. **Trigger**, **Toggle**- and **Mersenne-mode** operate at audio rates and can be used as Lo-Fi Amp, Distortion, Bit-Crushing effects with VCOs.

Probability knob is applied globally - rejected triggers can be recovered on the *utility output* (trINV mode) for chaining.

Trigger outputs 1 short pulse whenever input exceeds the threshold. it won't fire again until the input signal passes between -0.3 to 0.3 volt.

Gate length set to 0 (*Automatic*):

Gate will, unless probability intervenes, go high along with the input and pressed/held button and stay high until threshold is passed.

Gate length set above 0 (*100ms - 10sec*):

Gate performs autonomous and can't be interrupted. It goes high with an incoming signal for the set duration. Unless otherwise specified the rest of the module will still process incoming pulses. Via the context menu you can freeze the module while **Gate** is high / turn **Gate** to an *AHD-Envelope Attack-Hold-Decay* /

End of Gate & Start+End of Gate utility mode can make **Gate** a timed event trigger. Future updates will expand on that notion.

Its simplest function is as trigger to gate converter.

_context-menu:

__ turning re-trigger off globally will make Gate the centerpiece of Button and has many practical uses

__ set Rise and Fall (0-50%). At 50% Rise and Fall you will get a triangle wave and at lower rates an Attack-Hold-Decay function.

A | B are multi-mode operators.

Modes are set by Switch (*click to cycle, right-click to chose directly*):

_Toggle switches between A and B whenever the module is triggered. Both output a *constant voltage* set by the range & offset knobs (-10v to 10v) when active and 0v when inactive. Plug A and B output into the same input to switch between the two voltages. Toggle runs at audio rate and halves the incoming signals sample-rate: use probability to add fluctuations.

_RND bipolar independently generates random voltages at A and B outputs.

Range- & offset knobs set bounds: *bipolar range of 2 with an offset of 1 equals -1v to 3v as range for generated voltage (max -10v to 10v).*

_RND unipolar operates the same but in 0v to 10v.

context menu offers additional control:

__ **quantization** to semitones

__ dedicated **clock divider** for A and B respectively

__ dedicated **probability** for A and B respectively

__ switch between **uniform** and **normal distribution**

_Mersenne Twister

[MT19937 by Takuji Nishimura and Makoto Matsumoto](#)

Pseudo Random Number Generator that generates consistent streams of repeatable (fixed) number sequences based on a seed value. Each sequence can be repeated by resetting the generator every time, provided one knows the corresponding seed value.

Mersenne features a sequence mode in the context menu:

__set a seed (0-10000)

__activate sequence

__set the desired number of steps (2-128)

Same as RND.. range, offset, semitone, clock div and probability apply.

Changing the seed number needs a trigger/button to go into effect.

Mersenne sounds surprisingly musical and is a great source for pseudo-random note generation.

It can be driven at audio-rates and setting the range and offset have great effect while probability and clock division mess with signals sample-rate.

_Ramps & latching Ramps will generate a rising or falling linear voltage.

Times are dictated by Gate length while range and offset set the target (rising ramps) or starting (falling ramps) value. The lower the range the slower the ramp.

Regular ramps return to 0v after finishing, latching ramps remain at the final value until the next trigger is recieved. in the context menu you can set the ramps to rising or falling respectively. Use a ramp to time event triggers with comparators, drive phasor modules or use them as modulation sources.